

# CS 156

## Intro to AI



## Machine Learning: Naïve Bayes

These slides are based on the slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.  
The artwork is by Ketrina Yim

# Machine Learning

- ▶ Up until now: how to **use** a model to make optimal decisions
- ▶ Machine learning: how to **acquire** a model from **data / experience**
  - Learning parameters (e.g. probabilities)
  - Learning structure (e.g. BN graphs – where are the arcs?)
  - Learning hidden concepts (e.g. clustering)

# Machine Learning

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"When you're fundraising, it's AI

When you're hiring, it's ML

When you're implementing, it's linear regression

When you're debugging, it's printf()"

~Baron Schwartz

# Machine Learning

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- ▶ Today: model-based classification with Naïve Bayes
  - What is classification?
  - What model?
  - What is naïve about it?

# Classification



# Classification Tasks

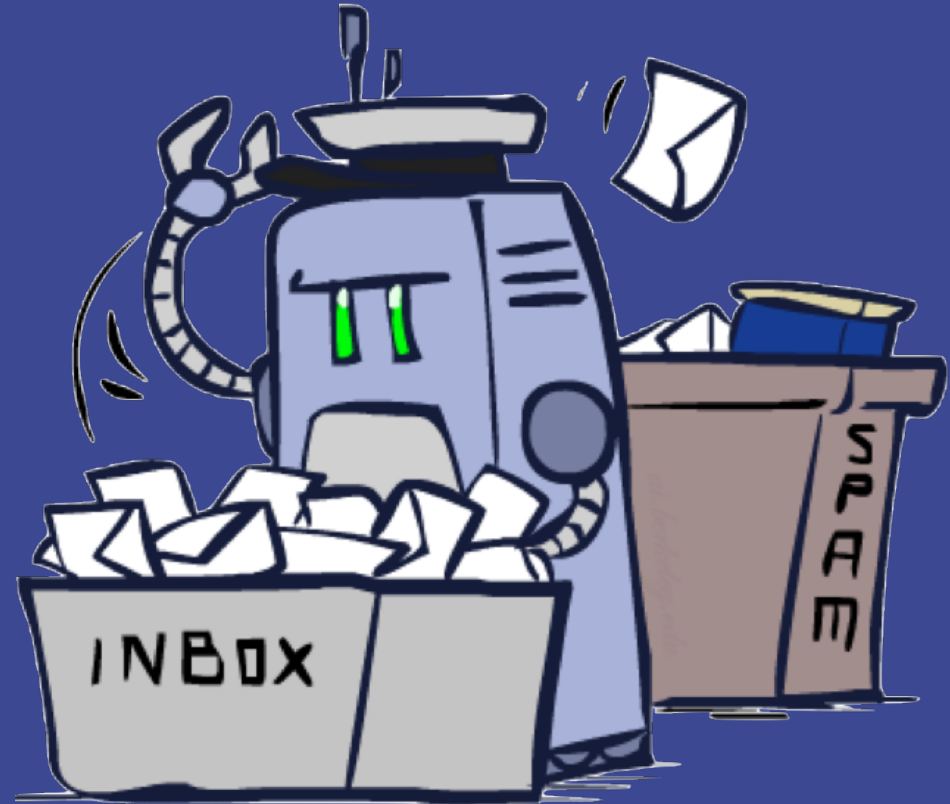
Classification: given **input**  $x$ , predict a **label (class)**  $y$

- ▶ Spam detection?  
input: email message, classes: spam / ham
- ▶ OCR - Optical Character Recognition?  
input: images, classes: characters
- ▶ Medical diagnosis?  
input: symptoms/test results, classes: diseases
- ▶ Automatic essay grading?  
input: document, classes: grades
- ▶ Fraud detection?  
input: account activity, classes: fraud / no fraud
- ... many more

Classification is an  
important **commercial**  
technology!

# Example: Spam Filter

- ▶ Input: an email
- ▶ Output: spam/ham



# Example: Spam Filter

## Setup:

- Get a large collection of example emails, each labeled "spam" or "ham"
- Note: someone has to hand label all this data!

Dear Sir.

First, I must solicit your confidence in this transaction, this is by virtue of its nature as being utterly confidential and TOP SECRET. ...

iClicker:

How would you label this message?

A. Ham

B. Spam



# Example: Spam Filter

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iClicker:

What is the most important word that informed your decision?

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iClicker:

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Ok, I know this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

# Example: Spam Filter

## Setup:

- Get a large collection of example emails, each labeled "spam" or "ham"
- Note: someone has to hand label all this data!
- Our goal: to learn to predict labels of **new, future emails**



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# Example: Spam Filter

► **Features:** The attributes used to make the ham / spam decision

- Words: FREE! SECRET
- Text Patterns: \$dd, CAPS
- Spelling errors
- Non-text: SenderInContacts
- ...



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# Example: Digit Recognition

- ▶ Input: images / pixel grids
- ▶ Output: a digit 0-9
- ▶ Setup:
  - Get a large collection of example images, each labeled with a digit
  - Note: someone has to hand label all this data!
  - Our goal: learn to predict labels of new, future digit images



0



1



2



1



??

# Example: Digit Recognition

- **Features:** The attributes used to make the digit decision
  - Pixels: (6,8)=ON
  - Shape Patterns: AspectRatio, NumLoops
  - ...



0



1



2



1

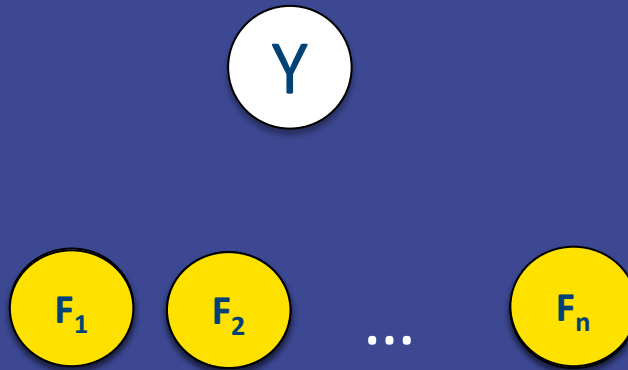


??

# Model-Based Classification

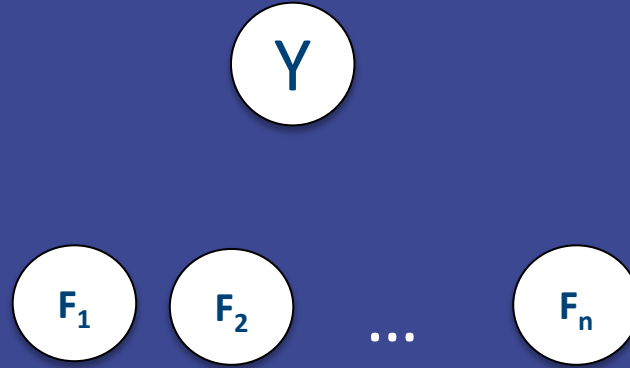
## Model-based approach

- ▶ Build a model (e.g. Bayes' net) where both the label  $Y$  and features  $F_1$  through  $F_n$  are random variables



- ▶ Instantiate any observed features
- ▶ Query for the distribution of the label conditioned on the observed features:  $P(Y|F)$

# Model-Based Classification

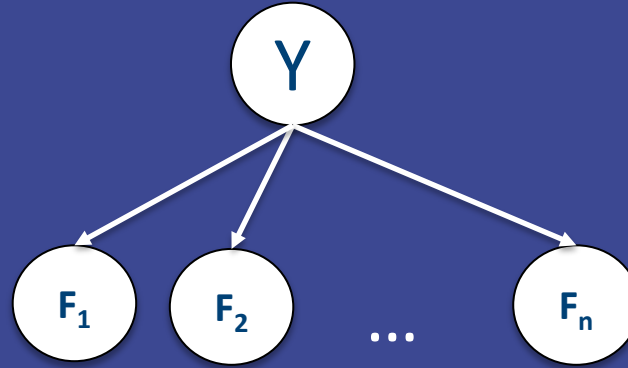


Challenges:

- ▶ What **structure** should the Bayes Net have? Where do we place the arcs?
- ▶ How should we learn its **parameters/probabilities**?



# Naïve Bayes?



What's naïve about it?

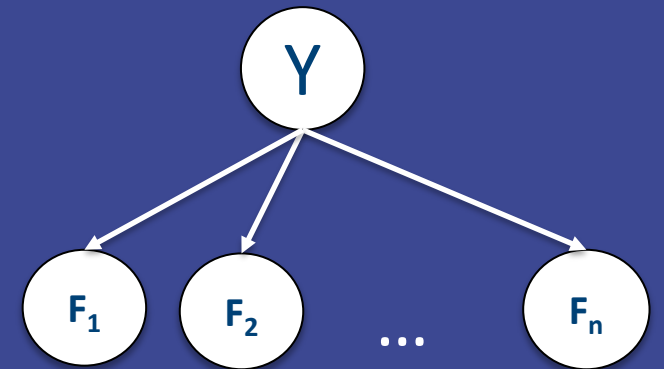
- Assume all **features are independent effects** of the label:

$$F_i \perp\!\!\!\perp F_j \mid Y$$

# Naïve Bayes for Digits

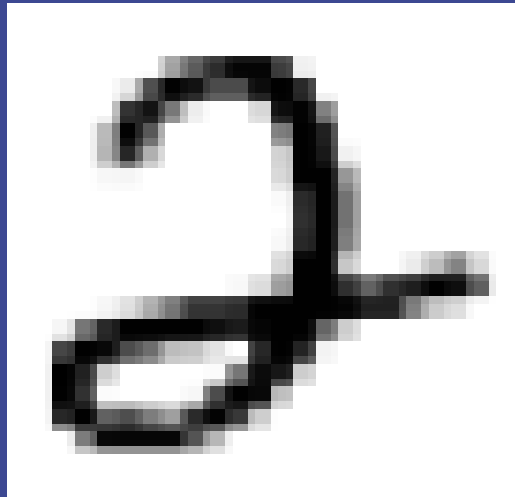
Simple digit recognition:

- ▶ One feature (variable)  $F_{ij}$  for each grid position  $\langle i, j \rangle$ /pixel
- ▶ Feature values are on / off, based on whether intensity is more or less than 0.5 in underlying image
- ▶ Each input maps to a **feature vector**



# Feature Vectors

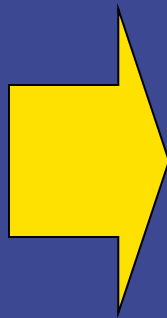
Data is represented as a **feature vector**


$$\left[ \begin{array}{l} \text{PIXEL-7,12} : 1 \\ \text{PIXEL-7,13} : 0 \\ \dots \end{array} \right]$$

# Feature Vectors

Data is represented as a **feature vector**

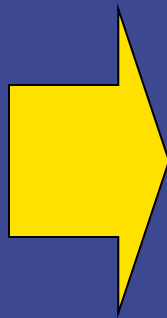
Hello,  
Do you want free  
printer  
cartridges? Why  
pay more when  
you can get them  
ABSOLUTELY FREE!  
Just ...


$$\begin{bmatrix} \# \text{ free} & : ? \\ \text{YOUR\_NAME} & : ? \\ \text{MISSPELLED} & : ? \\ \text{FROM\_FRIEND} & : ? \\ \dots & \end{bmatrix}$$

# Feature Vectors

Data is represented as a **feature vector**

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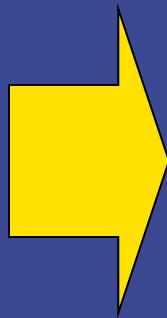


|             |    |
|-------------|----|
| # free      | :2 |
| YOUR_NAME   | :? |
| MISSPELLED  | :? |
| FROM_FRIEND | :? |
| ...         |    |

# Feature Vectors

Data is represented as a **feature vector**

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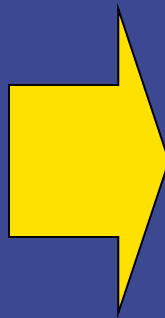


|             |    |
|-------------|----|
| # free      | :2 |
| YOUR_NAME   | :0 |
| MISSPELLED  | :? |
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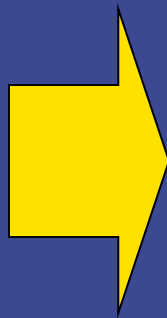


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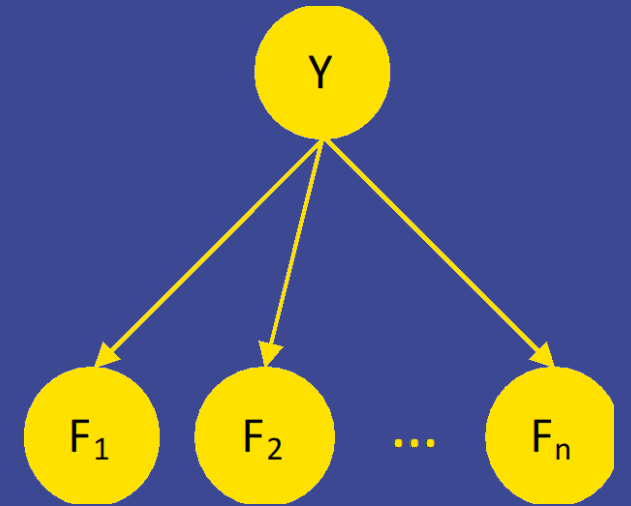


|             |    |
|-------------|----|
| # free      | :2 |
| YOUR_NAME   | :0 |
| MISSPELLED  | :2 |
| FROM_FRIEND | :0 |
| ...         |    |



# Recap: Naïve Bayes Classification

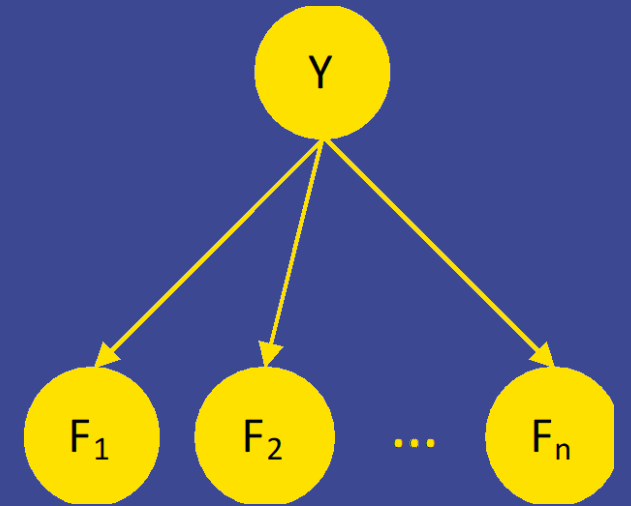
- ▶ Classification: given **inputs**  $x$ , predict **labels (classes)**  $y$
- ▶ **Features**: the input attributes used to make the classification
- ▶ **We** decide on the features
- ▶ Build a model (e.g. Bayes' net) where both the label and features are random variables
- ▶ **Naïve Bayes**: **Assume all features are independent effects of the label**
- ▶ Learn the probabilities from the data



# General Naïve Bayes

A general Naïve Bayes model:

- ▶  $P(Y, F_1, F_2, \dots F_n) = P(Y) P(F_1|Y)P(F_2|Y)\dots P(F_n|Y)$



- ▶ We only have to specify how each feature depends on the class
- ▶ Total number of parameters is *linear* in the number of features  $n$
- ▶ Model is very simplistic, but often works well anyway

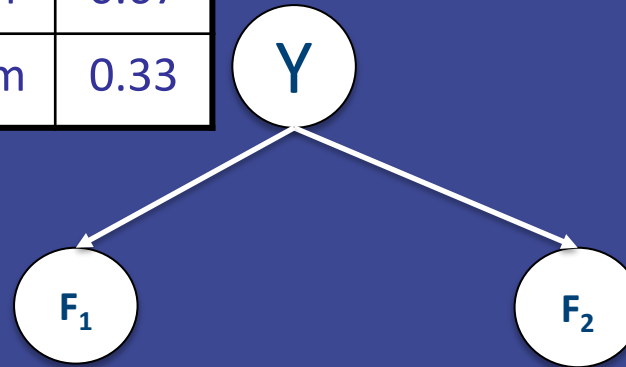
# General Naïve Bayes

What do we need in order to use Naïve Bayes?

1. **Inference method** (we saw this part in the previous couple of lectures)
  - Start with a bunch of probabilities:  $P(Y)$  and the  $P(F_i|Y)$  tables
  - Use probabilistic inference to compute  $P(Y|F_1...F_n)$  for all possible labels  $y$ .
  - Classify with the label  $y$  with the highest value of  $P(y|F_1...F_n)$ ..

| Y    | F1  | P(F1 Y) |
|------|-----|---------|
| ham  | +f1 | 0.01    |
| ham  | -f1 | 0.99    |
| spam | +f1 | 0.7     |
| spam | -f1 | 0.3     |

| Y    | P(Y) |
|------|------|
| ham  | 0.67 |
| spam | 0.33 |



| Y    | F2  | P(F2 Y) |
|------|-----|---------|
| ham  | +f2 | 0.9     |
| ham  | -f2 | 0.1     |
| spam | +f2 | 0.2     |
| spam | -f2 | 0.8     |

# General Naïve Bayes - Inference method

We have an email that exhibits +f1 and -f2 (contains word free and not from a friend)

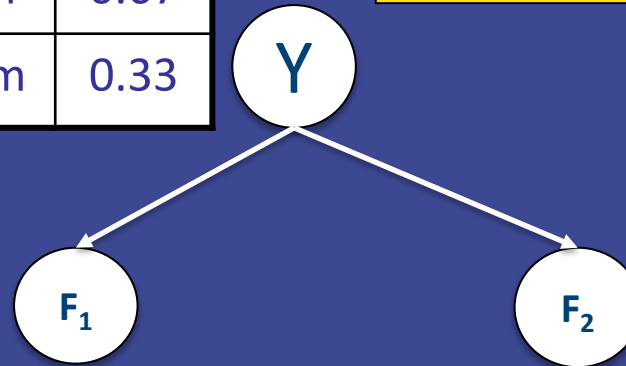
$$P(\text{ham} | +f1, -f2) = \frac{P(\text{ham}, +f1, -f2)}{P(+f1, -f2)}$$

iClicker: What is  $P(\text{ham}, +f1, -f2)$ ?

- A.  $P(\text{ham}).P(+f1).P(-f2)$
- B.  $P(\text{ham})+P(+f1)-P(-f2)$
- C.  $P(\text{ham})+P(+f1 | \text{ham})+P(-f2 | \text{ham})$
- D.  $P(\text{ham}).P(+f1 | \text{ham}).P(-f2 | \text{ham})$
- E.  $P(\text{ham}).P(+f1 | \text{ham}).P(-f2 | f1)$

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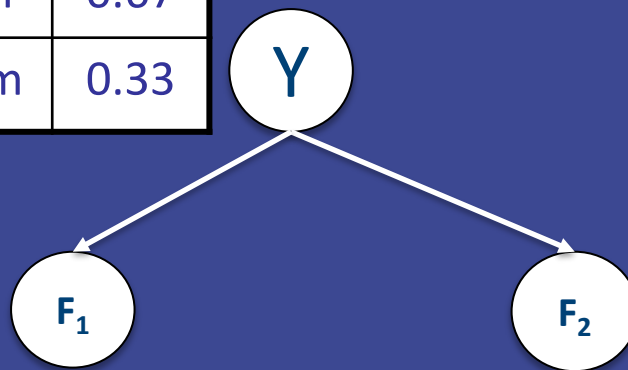
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|------|-----|---------|
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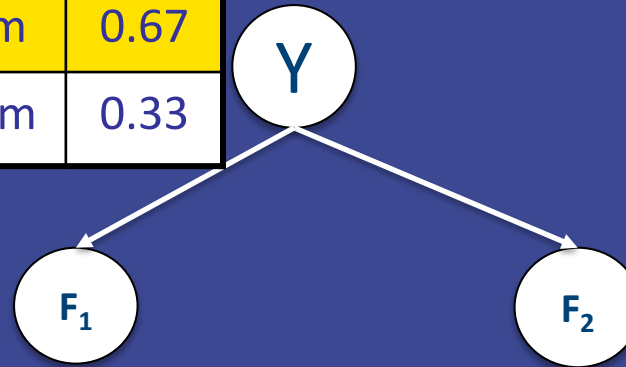
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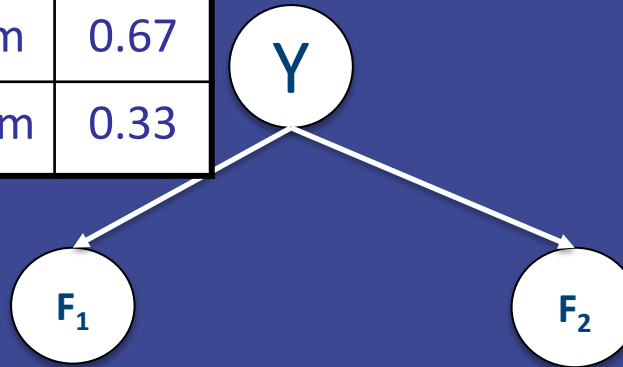
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$$P(\text{spam} | +f1, -f2) = \frac{P(\text{spam}, +f1, -f2)}{P(+f1, -f2)} = \frac{P(\text{spam}) \cdot P(+f1 | \text{spam}) \cdot P(-f2 | \text{spam})}{P(+f1, -f2)}$$

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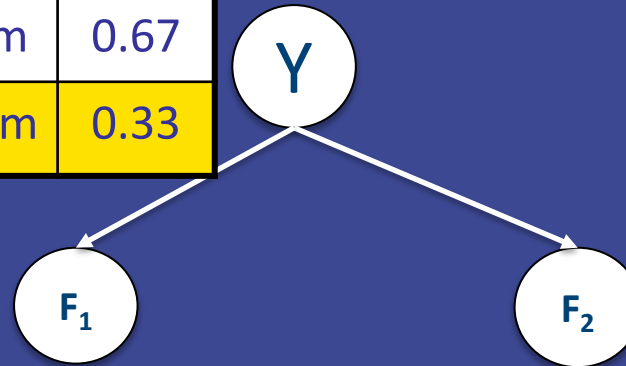
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$$P(\text{spam} | +f1, -f2) = \frac{P(\text{spam}, +f1, -f2)}{P(+f1, -f2)} = \frac{P(\text{spam}) \cdot P(+f1 | \text{spam}) \cdot P(-f2 | \text{spam})}{P(+f1, -f2)} = \frac{0.1848}{P(+f1, -f2)}$$

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iClicker: A naive Bayes classifier will classify our email as:

- A. Ham
- B. Spam

|      |     |      |
|------|-----|------|
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# General Naïve Bayes

What do we need in order to use Naïve Bayes?

## 1. Inference method

- ✓ Start with a bunch of probabilities:  $P(Y)$  and the  $P(F_i|Y)$  tables
- ✓ Use probabilistic inference to compute  $P(Y|F_1...F_n)$  for all possible labels  $y$ .
- ✓ Classify with the label  $y$  with the highest value of  $P(y|F_1...F_n)$ .

## 2. Estimates of local conditional probability tables

- $P(Y)$ , the prior over labels ( $P(\text{spam})$ ,  $P(\text{ham})$ )
- $P(F_i|Y)$  for each feature (evidence variable, for example  $P(\text{free}|\text{spam})$  ).
- These probabilities are collectively called the *parameters* of the model:  $\theta$
- Up until now, we assumed these probabilities appeared by magic, but...
- We can get them from **training data** counts: let's see how to do that next.

# Data

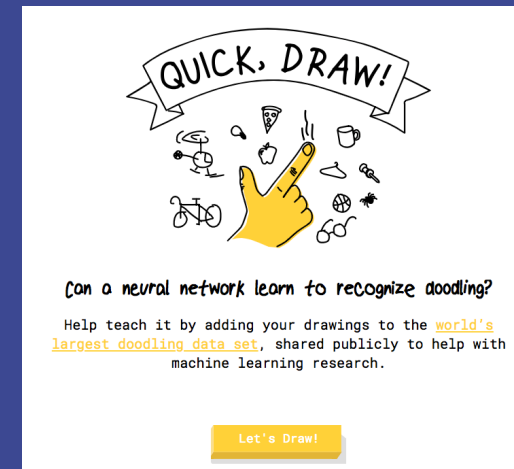
- Data: **labeled** instances, e.g. emails marked spam/ham



<https://xkcd.com/1838/>

# Data

- ▶ Where does the labeled data come from?
- ▶ Who labels it?
  - Public datasets (Kaggle)
  - Amazon Mechanical Turk
  - Games/Apps (Google Quick Draw, Duolingo)

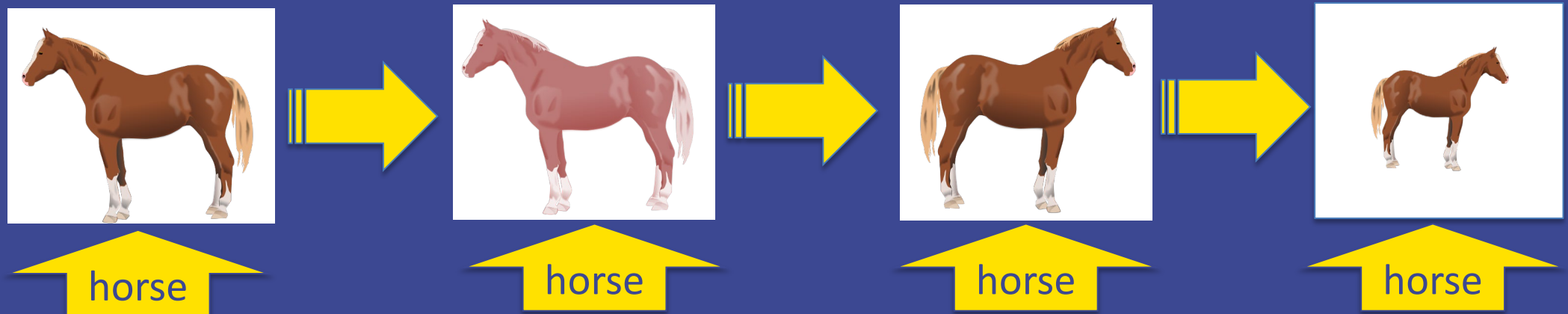


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- ▶ Not enough data? Data Augmentation!
  - Examples: images, language (Natural Language Processing)

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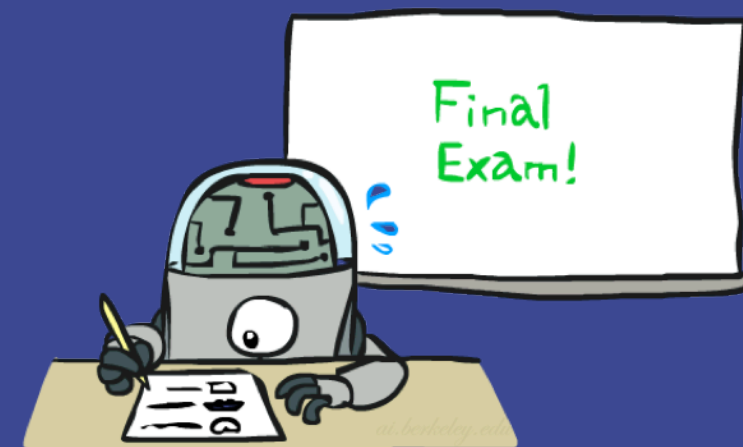
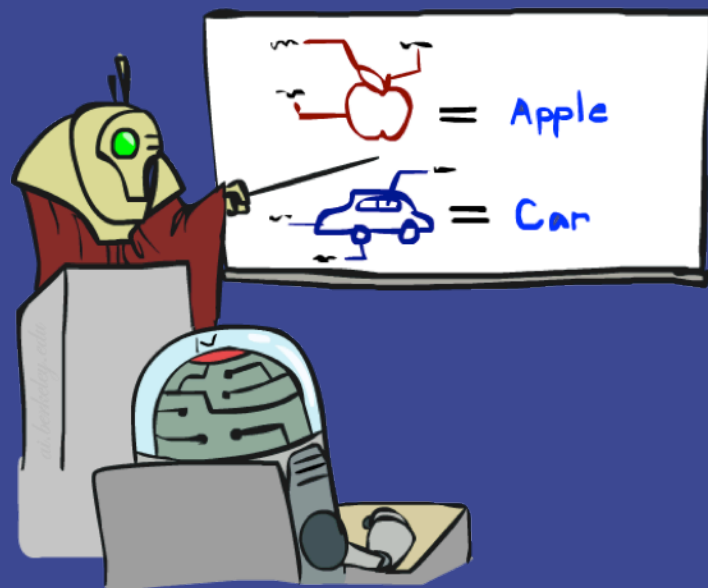


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- ▶ Not enough data? Data Augmentation!
  - Examples: images, language (Natural Language Processing)
- ▶ Important: data bias?



# Training and Testing



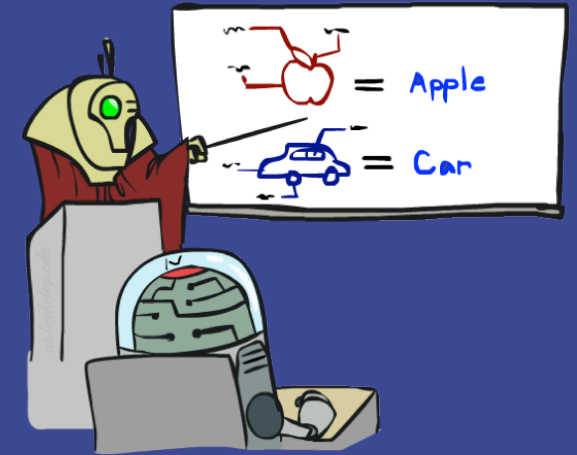
# Data

- ▶ We divide our data into 3 sets:
  - Training set (60%)
  - Held out set / validation (20%)
  - Test set (20%)

Training  
Data

Validation  
Data

Test  
Data



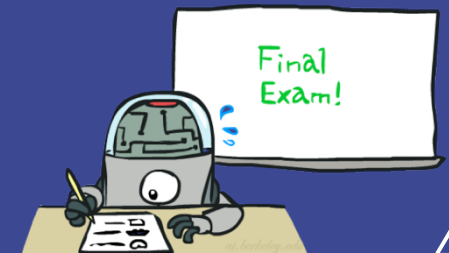
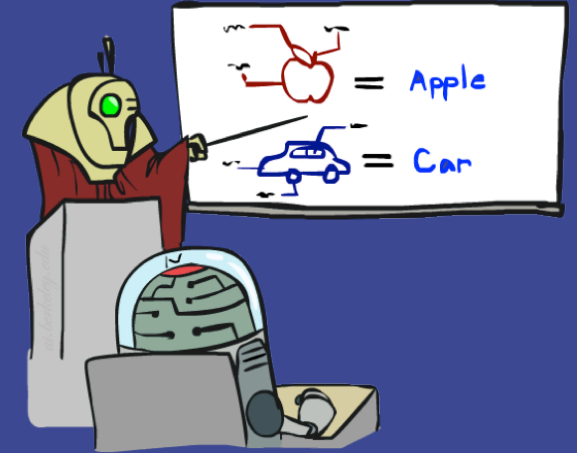
# Training and Testing

- ▶ Experimentation cycle
  - **Learn** parameters (model probabilities) from **training set**
  - **Tune** hyperparameters on **validation set**
  - Compute **accuracy** on **test set**
  - Very important: never "peek" at the test set!

Training  
Data

Validation  
Data

Test  
Data



# Training and Testing

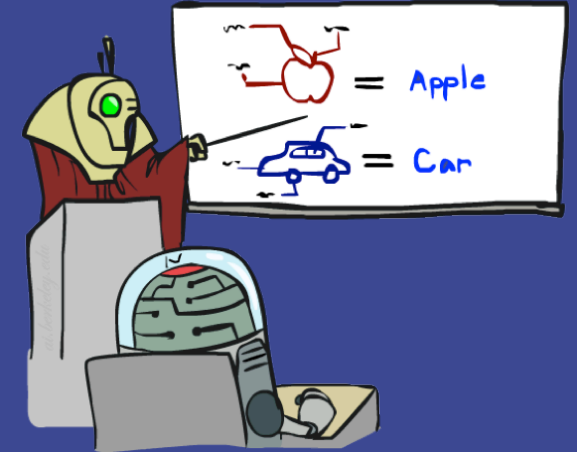
## ► Evaluation

- Accuracy: fraction of instances predicted correctly
- Risk and utility: some 'mistakes' are worse than others (classifying an important email as spam)

Training  
Data

Validation  
Data

Test  
Data



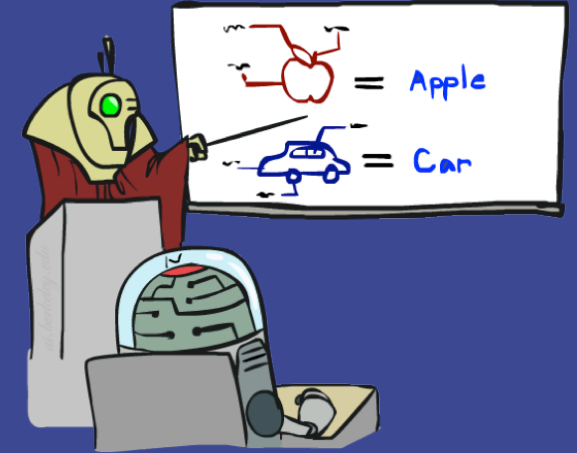
# Training and Testing

- ▶ Overfitting and generalization
  - We want a classifier which does well on **test data**
  - **Overfitting**: fitting the training data very closely, but not generalizing well
  - More on overfitting later

Training  
Data

Validation  
Data

Test  
Data



# iClicker: Supervised vs Unsupervised

Naïve Bayes is an example of a \_\_\_\_\_ learning algorithm.

- A. supervised
- B. unsupervised

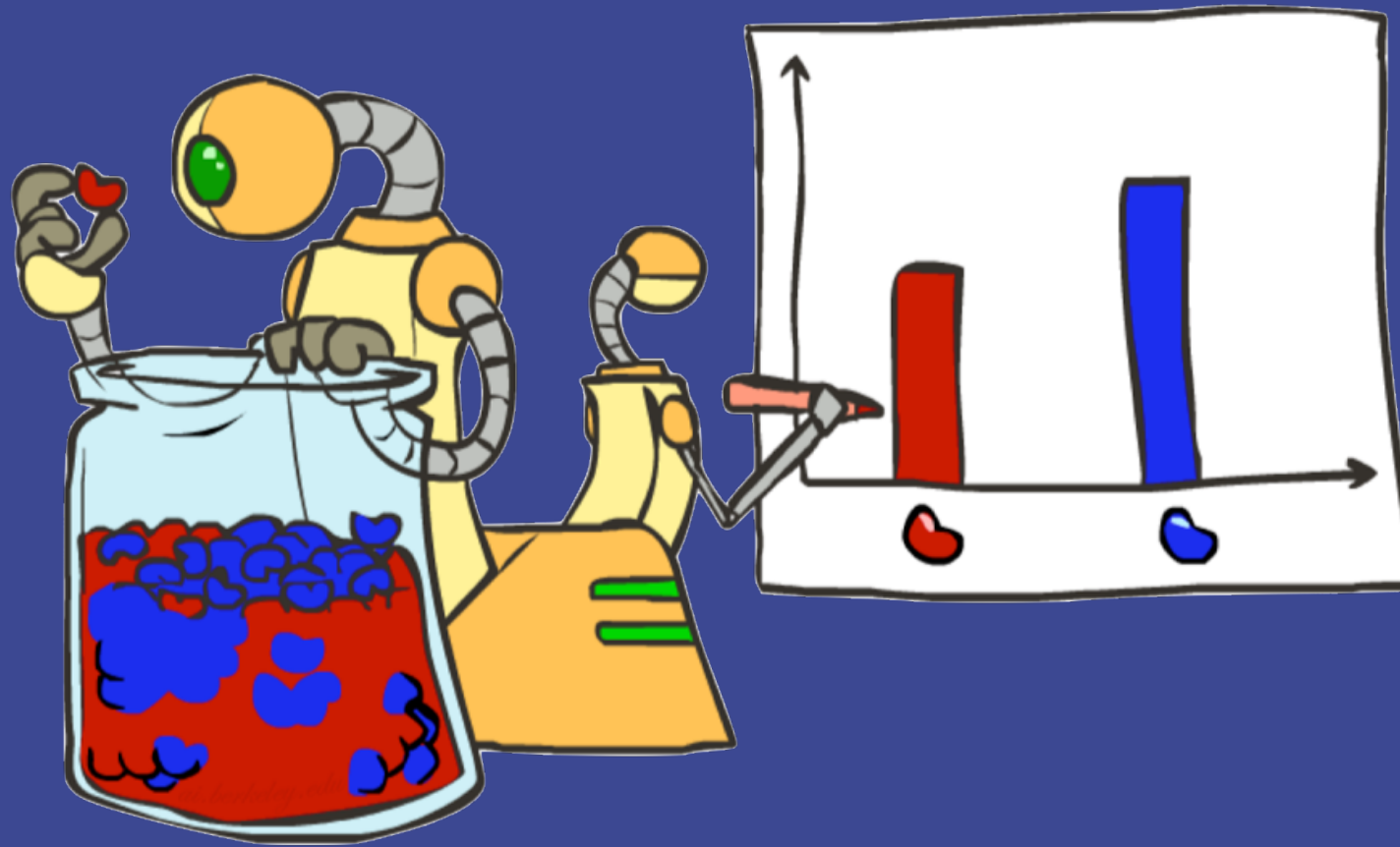
# Supervised vs Unsupervised

---

Naïve Bayes is an example of a supervised learning algorithm.

This means that we give the algorithm a data set in which the "right answers" are given.

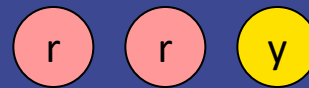
# Parameter Estimation





# Parameter Estimation

- ▶ Estimate the distribution of a random variable (**class or feature**)
- ▶ Empirically: using the training data (learning!)
  - For each outcome, look at the **empirical rate** of that outcome:
  - $P_{ML}(x) = \frac{count(x)}{total\ samples}$
  - $P_{ML}(r) = 2/3$
  - This is the **maximum likelihood** estimate.



# Example: Digit Recognition

$P(Y)$

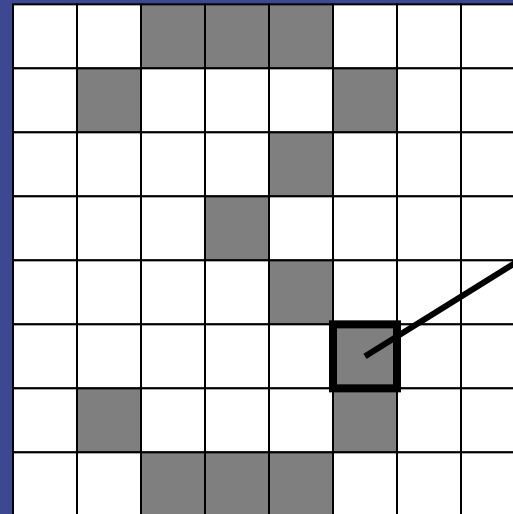
|   |      |
|---|------|
| 1 | 0.14 |
| 2 | 0.08 |
| 3 | 0.10 |
| 4 | 0.06 |
| 5 | 0.07 |
| 6 | 0.09 |
| 7 | 0.09 |
| 8 | 0.11 |
| 9 | 0.13 |
| 0 | 0.13 |

# Example: Digit Recognition

$P(Y)$

|   |      |
|---|------|
| 1 | 0.14 |
| 2 | 0.08 |
| 3 | 0.10 |
| 4 | 0.06 |
| 5 | 0.07 |
| 6 | 0.09 |
| 7 | 0.09 |
| 8 | 0.11 |
| 9 | 0.13 |
| 0 | 0.13 |

Training Label: 3



$P(F_{5,5} = \text{on} | Y)$

|   |      |
|---|------|
| 1 | 0.05 |
| 2 | 0.01 |
| 3 | 0.90 |
| 4 | 0.80 |
| 5 | 0.90 |
| 6 | 0.90 |
| 7 | 0.25 |
| 8 | 0.85 |
| 9 | 0.60 |
| 0 | 0.80 |

# Example: Spam Filtering

---

- ▶ Bag-of-words model:
  - Simplified representation commonly used in Natural Language Processing
  - Message is modeled as an **unordered** collection (bag) of words
  - Grammar is also ignored

# Example: Spam Filtering

- What are the parameters?

$P(Y)$

ham : 0.66  
spam: 0.33

$P(W|\text{spam})$

the : 0.0156  
to : 0.0153  
and : 0.0115  
of : 0.0095  
you : 0.0093  
a : 0.0086  
with: 0.0080  
from: 0.0075  
...

$P(W|\text{ham})$

the : 0.0210  
to : 0.0133  
of : 0.0119  
2002: 0.0110  
with: 0.0108  
from: 0.0107  
and : 0.0105  
a : 0.0100  
...

# Example: Spam Filtering

| Word    | P(w spam) | P(w ham) | Tot Spam | Tot Ham |
|---------|-----------|----------|----------|---------|
| (prior) | 0.33333   | 0.66666  | 0.33     | 0.67    |
| Gary    | 0.00002   | 0.00021  |          |         |
| would   | 0.00069   | 0.00084  | 0.04     | 0.96    |
| you     | 0.00881   | 0.00304  | 0.10     | 0.90    |
| like    | 0.00086   | 0.00083  | 0.11     | 0.89    |
| to      | 0.01517   | 0.01339  | 0.12     | 0.88    |
| lose    | 0.00008   | 0.00002  | 0.35     | 0.65    |
| weight  | 0.00016   | 0.00002  | 0.81     | 0.19    |
| while   | 0.00027   | 0.00027  | 0.81     | 0.19    |
| you     | 0.00881   | 0.00304  | 0.93     | 0.07    |
| sleep   | 0.00006   | 0.00001  | 0.99     | 0.01    |

$$P(\text{spam} | \text{Gary}) = \frac{P(\text{spam}) P(\text{Gary} | \text{spam})}{P(\text{Gary})}$$

$$\frac{0.33 \times 0.00002}{P(\text{Gary})}$$

$$P(\text{ham} | \text{Gary}) = \frac{P(\text{ham}) P(\text{Gary} | \text{ham})}{P(\text{Gary})}$$

$$\frac{0.66 \times 0.00021}{P(\text{Gary})}$$

$$Z = 0.33 \times 0.00002 + 0.66 \times 0.00021$$

$$P(\text{spam} | \text{Gary}) = \frac{0.33 \times 0.00002}{Z}$$

$$P(\text{spam} | w) = 99\%$$

# Reminders

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- ▶ Next Tuesday is a holiday: no class!
- ▶ Homework 6 due on Thursday
- ▶ Question of the Week on Thursday